



chemical changes

Name: _____

Class: _____

Date: _____

Time:

Marks: **75 marks**

Comments:

1.

This question is about displacement reactions.

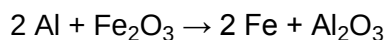
- (a) The displacement reaction between aluminium and iron oxide has a high activation energy.

What is meant by 'activation energy'?

(1)

- (b) A mixture contains 1.00 kg of aluminium and 3.00 kg of iron oxide.

The equation for the reaction is:



Show that aluminium is the limiting reactant.

Relative atomic masses (A_r): O = 16 Al = 27 Fe = 56

(4)

Magnesium displaces zinc from zinc sulfate solution.

- (c) Complete the ionic equation for the reaction.

You should include state symbols.



(2)

- (d) Explain why the reaction between magnesium atoms and zinc ions is both oxidation and reduction.

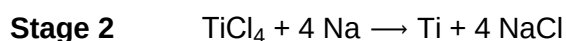
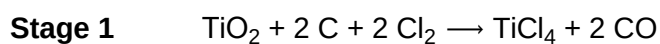
(2)

(Total 9 marks)

2.

Titanium is a transition metal.

Titanium is extracted from titanium dioxide in a two-stage industrial process.



- (a) Suggest **one** hazard associated with **Stage 1**.

(1)

- (b) Water must be kept away from the reaction in **Stage 2**.

Give **one** reason why it would be hazardous if water came into contact with sodium.

(1)

- (c) Suggest why the reaction in **Stage 2** is carried out in an atmosphere of argon and **not** in air.

(2)

- (d) Titanium chloride is a liquid at room temperature.

Explain why you would **not** expect titanium chloride to be a liquid at room temperature.

(3)

In **Stage 2**, sodium displaces titanium from titanium chloride.

- (e) Sodium atoms are oxidised to sodium ions in this reaction.

Why is this an oxidation reaction?

(1)

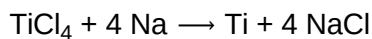
- (f) Complete the half equation for the oxidation reaction.



(1)

- (g) In Stage 2, 40 kg of titanium chloride was added to 20 kg of sodium.

The equation for the reaction is:



Relative atomic masses (A_r): Na = 23 Cl = 35.5 Ti = 48

Explain why titanium chloride is the limiting reactant.

You **must** show your working.

(4)

- (h) For a **Stage 2** reaction the percentage yield was 92.3%

The theoretical maximum mass of titanium produced in this batch was 13.5 kg.

Calculate the actual mass of titanium produced.

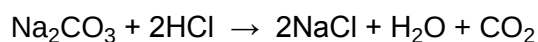
Mass of titanium = _____ kg

(2)

(Total 15 marks)

3.

Sodium carbonate reacts with dilute hydrochloric acid:

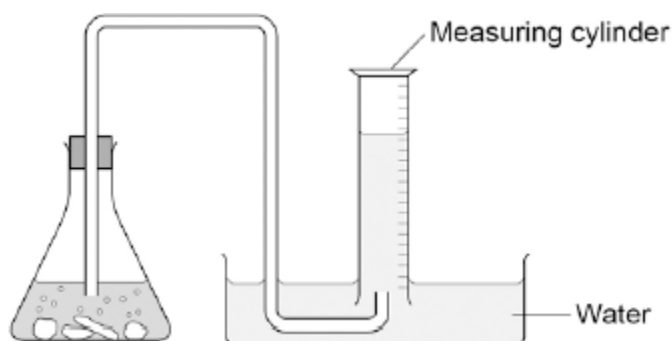


A student investigated the volume of carbon dioxide produced when different masses of sodium carbonate were reacted with dilute hydrochloric acid.

This is the method used.

1. Place a known mass of sodium carbonate in a conical flask.
2. Measure 10 cm³ of dilute hydrochloric acid using a measuring cylinder.
3. Pour the acid into the conical flask.
4. Place a bung in the flask and collect the gas until the reaction is complete.

(a) The student set up the apparatus as shown in the figure below.



Identify the error in the way the student set up the apparatus.

Describe what would happen if the student used the apparatus shown.

(2)

- (b) The student corrected the error.

The student's results are shown in the table below.

Mass of sodium carbonate in g	Volume of carbon dioxide gas in cm ³
0.07	16.0
0.12	27.5
0.23	52.0
0.29	12.5
0.34	77.0
0.54	95.0
0.59	95.0
0.65	95.0

The result for 0.29 g of sodium carbonate is anomalous.

Suggest what may have happened to cause this anomalous result.

(1)

- (c) Why does the volume of carbon dioxide collected stop increasing at 95.0 cm³?

(1)

- (d) What further work could the student do to be more certain about the minimum mass of sodium carbonate needed to produce 95.0 cm³ of carbon dioxide?

(1)

- (e) The carbon dioxide was collected at room temperature and pressure.
The volume of one mole of any gas at room temperature and pressure is 24.0 dm^3 .

How many moles of carbon dioxide is 95.0 cm^3 ?

Give your answer in three significant figures.

_____ mol

(2)

- (f) Suggest **one** improvement that could be made to the apparatus used that would give more accurate results.

Give a reason for your answer.

(2)

- (g) One student said that the results of the experiment were wrong because the first few bubbles of gas collected were air.

A second student said this would make no difference to the results.

Explain why the second student was correct.

(2)

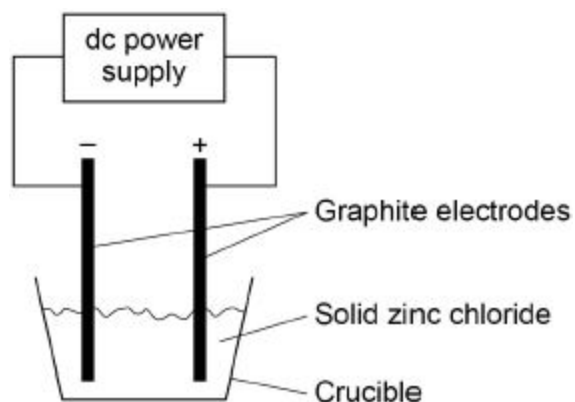
(Total 11 marks)

4.

A student investigated the electrolysis of different substances.

Figure 1 shows the apparatus.

Figure 1



- (a) Explain why electrolysis would not take place in the apparatus shown in **Figure 1**.

(2)

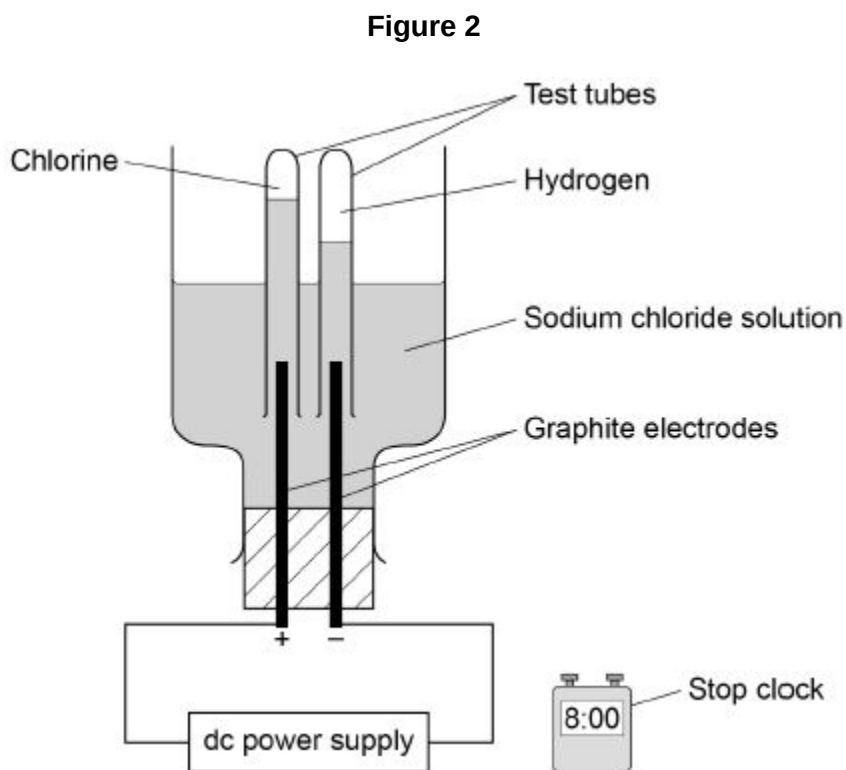
- (b) Explain why graphite conducts electricity.

Answer in terms of the structure and bonding in graphite.

(3)

The student investigated how the volume of gases produced changes with time in the electrolysis of sodium chloride solution.

Figure 2 shows the apparatus.



(c) The student made an error in selecting the apparatus for this investigation.

How should the apparatus be changed?

Give **one** reason for your answer.

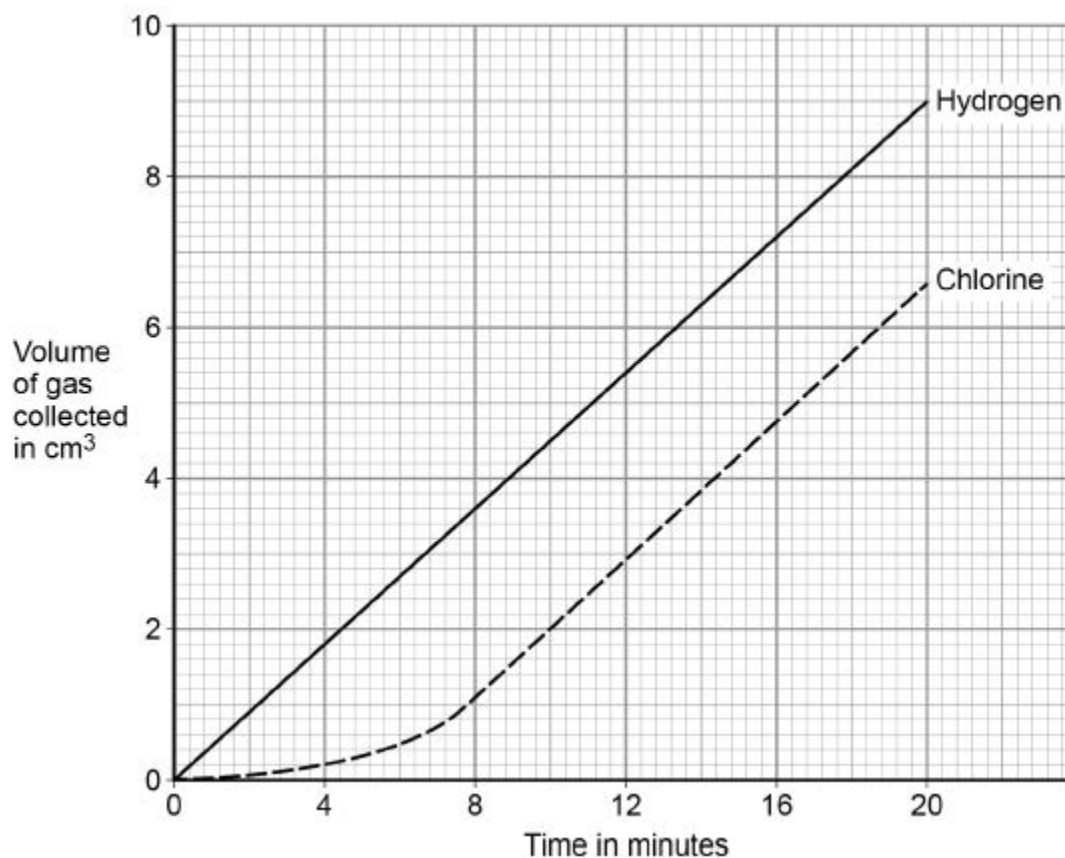
(2)

Another student used the correct apparatus.

This student measured the volumes of gases collected every minute for 20 minutes.

Figure 3 shows the student's results.

Figure 3



(d) Describe the trends shown in the results.

Use values from **Figure 3**.

(3)

- (e) The number of moles of each gas produced at the electrodes is the same.

No gas escapes from the apparatus.

Suggest **one** reason for the difference in volume of each gas collected.

(1)

- (f) Calculate the amount in moles of chlorine collected after 20 minutes.

Use **Figure 3**.

The volume of one mole of any gas at room temperature and pressure is 24.0 dm^3

Give your answer in standard form.

Moles of chlorine = _____ mol

(3)

(Total 14 marks)

5.

This question is about potassium.

- (a) Humphrey Davy was a professor of chemistry.

In 1807 Davy did an electrolysis experiment to produce potassium.

- (i) Davy first tried to electrolyse a solid potassium salt to produce potassium.

Explain why this electrolysis did **not** work.

(2)

- (ii) Humphrey Davy was the first person to produce potassium.

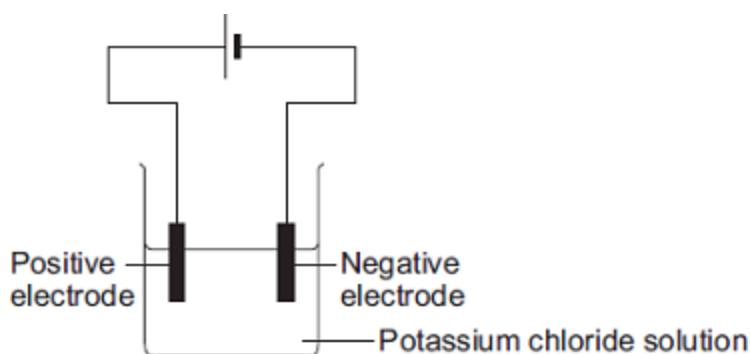
Humphrey Davy's experiment to produce this new element was quickly accepted by other scientists.

Suggest why.

(1)

- (b) A student dissolved some potassium chloride in water. The student tried to electrolyse the potassium chloride solution to produce potassium.

The apparatus the student used is shown in the diagram.



The student expected to see potassium metal at the negative electrode, but instead saw bubbles of a gas.

- Name the gas produced at the negative electrode.
- Explain why this gas was produced at the negative electrode **and** why potassium was not produced.

The reactivity series of metals on the Chemistry Data Sheet may help you to answer this question.

(3)

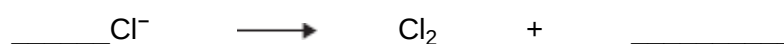
(c) The student tried to electrolyse molten potassium chloride to produce potassium.

(i) Potassium metal was produced at the negative electrode.

Describe how potassium atoms are formed from potassium ions.

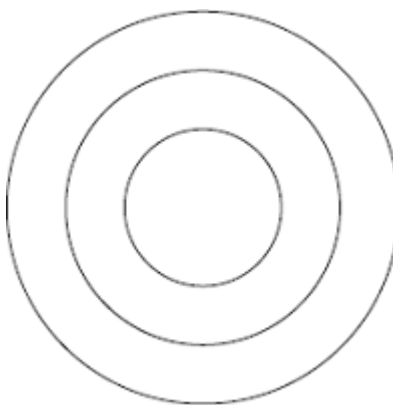
(2)

(ii) Complete and balance the equation for the reaction at the positive electrode.



(1)

(iii) Complete the diagram to show the electronic structure of a chloride ion (Cl^-).



(1)

(Total 10 marks)

6.

This question is about electrolysis.

Aluminium is produced by electrolysis of a molten mixture of aluminium oxide and cryolite.

(a) Explain why a mixture is used as the electrolyte instead of using only aluminium oxide.

(2)

(b) What happens at the negative electrode during the production of aluminium?

Tick (✓) **one** box.

Aluminium atoms gain electrons.

☐

Aluminium atoms lose electrons.

☐

Aluminium ions gain electrons.

☐

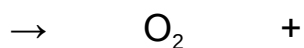
Aluminium ions lose electrons.

☐

(1)

(c) Oxygen is produced at the positive electrode.

Complete the balanced half-equation for the process at the positive electrode.

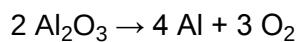


(2)

(d) Explain why the positive electrode must be continually replaced.

(3)

- (e) The overall equation for the electrolysis of aluminium oxide is:



Calculate the mass of oxygen produced when 2000 kg of aluminium oxide is completely electrolysed.

Relative atomic masses (A_r): O = 16 Al = 27

Mass of oxygen = _____ kg

(4)

Sodium metal and chlorine gas are produced by the electrolysis of molten sodium chloride.

- (f) Explain why sodium chloride solution **cannot** be used as the electrolyte to produce sodium metal.

(2)

- (g) Calculate the volume of 150 kg of chlorine gas at room temperature and pressure.

The volume of one mole of any gas at room temperature and pressure is 24.0 dm^3

Relative formula mass (M_r): $\text{Cl}_2 = 71$

Volume = _____ dm^3

(2)

(Total 16 marks)

Mark schemes

1.

- (a) the (minimum) energy needed for particles to react
or
the (minimum) energy needed for a reaction to occur
allow the (minimum) energy needed to start a reaction

1

- (b) (M_r of Fe_2O_3 =) 160

1

$$(\text{moles } \text{Fe}_2\text{O}_3 = \frac{3000}{160} =)$$

18.75 (mol)

allow correct use of incorrectly calculated M_r

1

$$(\text{moles Al} = \frac{1000}{27} =) 37.0 \text{ (mol)}$$

allow 37.037037 (mol) correctly rounded to at least 2 significant figures

*if both MP2 and MP3 are not awarded allow **1** mark for 0.01875 mol Fe_2O_3 **and** 0.037 mol Al*

1

(aluminium is limiting because)

37.0 mol is less than the ($2 \times 18.75 =$) 37.5 mol (aluminium needed)

or

iron oxide is in excess because 18.75 mol is more than the ($\frac{37.0}{2} =$) 18.5 mol (iron oxide needed)

allow correct use of incorrect number of moles from steps 2 and/or 3

alternative approaches:

approach 1:

(finding required mass of aluminium by moles method)

$$(M_r \text{ of Fe}_2\text{O}_3 =) 160 \text{ (1)}$$

$$(\text{moles Fe}_2\text{O}_3 = \frac{3000}{160} =)$$

$$18.75 \text{ (mol) (1)}$$

allow correct use of incorrectly calculated M_r

$$(\text{moles Al needed} = 18.75 \times 2 =) 37.5 \text{ (mol)}$$

and

$$(\text{mass Al needed} = 37.5 \times 27 =) 1012.5 \text{ (g) or } 1.0125 \text{ kg (1)}$$

allow correct use of incorrectly calculated moles of iron oxide

allow correct use of incorrectly calculated moles of aluminium needed

$$(\text{so) } 1.00 \text{ kg of aluminium is not enough (1)}$$

dependent on calculated mass of aluminium needed being greater than 1.00 (kg)

approach 2:

(finding required mass of aluminium by proportion method)

$$(M_r \text{ of Fe}_2\text{O}_3 =) 160 \text{ (1)}$$

$$(3.00 \text{ kg Fe}_2\text{O}_3 \text{ needs})$$

$$\frac{3.00}{160} \times 2 \times 27 \text{ (kg Al) (1)}$$

allow correct use of incorrectly calculated M_r

$$(=) 1.0125 \text{ (kg) (1)}$$

$$(\text{so) } 1.00 \text{ kg of aluminium is not enough (1)}$$

dependent on calculated mass of aluminium needed being greater than 1.00 (kg)

alternative approaches:

approach 3:
(finding required mass of iron oxide by moles method)

M_r of $\text{Fe}_2\text{O}_3 \Rightarrow 160$ (1)

(moles Al = $\frac{1000}{27} \Rightarrow 37.0$ (mol) (1)

allow 37.037037 (mol) correctly rounded to at least 2 significant figures

(moles Fe_2O_3 needed) = $\frac{37.0}{2} = 18.5$ (mol)

and

(mass Fe_2O_3 needed = $18.5 \times 160 \Rightarrow 2960$ (g) **or** 2.96 (kg) (1)

allow correct use of incorrectly calculated moles of aluminium

allow correct use of incorrectly calculated moles of iron oxide needed

allow correct use of incorrectly calculated M_r

(so) 3.00 kg of iron oxide is an excess (1)

dependent on calculated mass of iron oxide needed being less than 3.00 (kg)

approach 4:
(finding required mass of iron oxide by proportion method)

(M_r of $\text{Fe}_2\text{O}_3 \Rightarrow 160$ (1)

(1.00 kg Al needs) $\frac{1.00}{2 \times 27}$ (kg Fe_2O_3) (1)

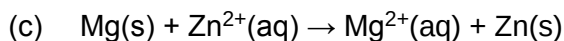
allow correct use of incorrectly calculated M_r

(=) 2.96 (kg) (1)

(so) 3.00 kg of iron oxide is an excess (1)

dependent on calculated mass of iron oxide needed being less than 3.00 (kg)

1



allow multiples

allow 1 mark for $\text{Mg}^{2+} + \text{Zn}$ with missing or incorrect state symbols

2

(d) magnesium (atoms) are oxidised because they lose electrons

1

(and) zinc (ions) are reduced because they gain electrons

*if no other marks awarded allow 1 mark for magnesium
(atoms) lose electrons and zinc (ions) gain electrons 1*

1

[9]

2.

(a) chlorine is toxic

allow carbon monoxide is toxic

allow poisonous for toxic

ignore harmful / deadly / dangerous

allow a poisonous gas is used / produced

allow titanium chloride is corrosive

1

(b) any **one** from:

- very exothermic reaction

allow explosive

allow violent reaction

ignore vigorous reaction

ignore sodium is very reactive

- produces a corrosive solution

allow caustic for corrosive

ignore alkaline

- produces hydrogen, which is explosive / flammable

allow flames produced

ignore sodium burns

1

(c) argon is unreactive / inert

*allow argon will not react (with reactants / products /
elements)*

1

oxygen (from air) would react with sodium / titanium

or

water vapour (from air) would react with sodium / titanium

*allow elements / reactants / products for sodium /
titanium*

1

(d) metal chlorides are usually ionic

allow titanium chloride is ionic

1

(so)(metal chlorides) are solid at room temperature

or

(so)(metal chlorides) have high melting points

allow titanium chloride for metal chlorides

1

(because) they have strong (electrostatic) forces between the ions

ignore strong ionic bonds

or

(but) must be a small molecule or covalent

allow molecular

1

allow alternative approach:

*titanium chloride must be covalent **or** has small molecules (1)*

with weak forces between molecules

*do **not** accept bonds unless intermolecular bonds(1)*

(but) metal chlorides are usually ionic (1)

(e) sodium (atoms) lose electrons

*do **not** accept references to oxygen*

1

(f) $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$

*do **not** accept e for e^-*

1

(g) (M_r of TiCl_4 =) 190

$$(\text{moles Na} = \frac{20\,000}{23} =) 870 \text{ (mol) } *$$

1

$$(\text{moles TiCl}_4 = \frac{40\,000}{190} =) 211 \text{ (mol) } *$$

1

allow 1 mark for 0.870 mol Na **and 0.211 mol TiCl_4*

allow use of incorrectly calculated M_r from step 1

either

(sodium is in excess because) 870 mol Na is more than the 844 mol needed

or

(because) 211 mol TiCl_4 is less than the 217.5 mol needed

the mark is for correct application of the factor of 4

other correct reasoning showing, with values of moles or mass, an excess of sodium or insufficient TiCl_4 is acceptable

allow use of incorrect number of moles from steps 2 and / or 3

1

alternative approaches:

approach 1:

(M_r of TiCl_4 =) 190(1)

(40 kg TiCl_4 needs)

$$\frac{40}{190} \times 4 \times 23 \text{ (kg Na) (1)}$$

(=) 19.4 (kg) (1)

so 20 kg is an excess (1)

approach 2:

(M_r of TiCl_4 =) 190(1)

(20 kg Na needs)

$$\frac{20}{4 \times 23} \times 190 \text{ (kg TiCl}_4\text{) (1)}$$

(=) 41.3 (kg) (1)

so 40 kg is not enough (1)

(h) (actual mass $\Rightarrow \frac{92.3}{100} \times 13.5$

or

(actual mass $\Rightarrow 0.923 \times 13.5$

1

$= 12.5 \text{ (kg)}$

allow 12 / 12.46 / 12.461 / 12.4605 (kg)

1

an answer 12.5 (kg) scores 2 marks

[15]

3.

(a) (delivery) tube sticks into the acid

1

the acid would go into the water **or** the acid would leave the flask or go up the delivery tube

ignore no gas collected

1

(b) any **one** from:

- bung not put in firmly / properly
- gas lost before bung put in
- leak from tube

1

(c) all of the acid has reacted

1

(d) take more readings in range 0.34 g to 0.54 g

1

take more readings is insufficient

ignore repeat

(e) $\frac{95}{24000}$

1

0.00396

or

3.96×10^{-3}

1

accept 0.00396 or 3.96×10^{-3} with no working shown for 2 marks

(f) use a pipette / burette to measure the acid

1

because it is more accurate volume than a measuring cylinder

or

greater precision than a measuring cylinder

or

use a gas syringe to collect the gas

so it will not dissolve in water

or

use a flask with a divider

accept description of tube suspended inside flask

so no gas escapes when bung removed

1

(g) they should be collected because carbon dioxide is left in flask at end

1

and it has the same volume as the air collected / displaced

1

[11]

4.

(a) solid (zinc chloride) does not conduct (electricity)

or

zinc chloride needs to be in solution **or** molten

allow liquid / aqueous

1

(because) ions cannot move in the solid

or

(as) ions can (only) move in liquid / solution

*do **not** accept references to movement of electrons in zinc chloride*

1

(b) each carbon / atom forms 3 (covalent) bonds

1

one electron per carbon / atom is delocalised

1

(so) these electrons carry charge through the graphite

or

(so) these electrons move through the structure

ignore carry current / electricity

1

*if no other mark scored, allow **1** mark for delocalised / free electrons*

allow free electrons for delocalised electrons

- (c) use measuring cylinders (instead of test tubes)

allow use burettes

allow use (gas) syringes

allow Hoffmann voltameter

1

(because) test tubes cannot measure volume

or

(because) test tubes have no graduations / scale

allow (so that) volume can be measured

1

- (d) any **three** from:

- the volume of hydrogen collected is directly proportional to the time
allow the (volume of) hydrogen is collected at a constant / steady rate
- the rate of collection of hydrogen is 0.45 (cm³/min)
- up to 8 minutes chlorine is collected at an increasing rate
allow any value from 6 to 8 minutes
allow initially chlorine is collected at an increasing rate
- after 8 minutes the rate of collection of chlorine is the same as that of hydrogen
allow any value from 6 to 8 minutes

or

after 8 minutes the rate of collection of chlorine is 0.45 (cm³/min)

allow after 8 minutes the (volume of) chlorine is collected at a constant / steady rate

if neither bullet point 3 nor bullet point 4 is awarded allow 1 mark for chlorine is collected slowly up to 8 minutes and then more quickly

allow any value from 6 to 8 minutes

3

- (e) chlorine reacts with water

or

chlorine dissolves (in the solution).

1

(f) $(\text{volume} =) \frac{6.6}{1000} \text{ (dm}^3\text{)}$

or 0.0066 (dm³)

allow 6.5 (cm³) for 6.6 (cm³)

1

$(\text{moles} =) \frac{0.0066}{24}$

allow use of incorrect volume from step 1

1

$= 2.75 \times 10^{-4} \text{ (mol)}$

allow $2.8 \times 10^{-4} \text{ (mol)}$

allow answer from incorrect calculation given in standard form

alternative approach for marking points 1 and 2

$24 \text{ dm}^3 = 24\,000 \text{ cm}^3 \text{ (1)}$

$(\text{moles} =) \frac{6.6}{24\,000} \text{ (1)}$

1

*an answer of $2.75 \times 10^{-4} \text{ (mol)}$ or $2.8 \times 10^{-4} \text{ (mol)}$ scores **3** marks*

*an answer of 0.000275 / 0.00028 / 2.75×10^{-1} / $2.8 \times 10^{-1} \text{ (mol)}$ / scores **2** marks*

*an incorrect answer for one step does **not** prevent allocation of marks for subsequent steps*

[10]

5.

(a) (i) current / charge couldn't flow

allow could not conduct (electricity)

1

because the ions / particles couldn't move

*do **not** accept electrons/ molecules / atoms*

or

(salt) needs to be molten / (1) dissolved (to conduct electricity)

so that the ions / particles can move (1)

*do **not** accept electrons / molecules / atoms*

1

(ii) he had status
*accept he had authority **or** experience*

or

he had evidence / proof
accept the experiment could be repeated

1

(b) hydrogen / H₂
*do **not** allow hydrogen ions*

1

the ions are positive
accept because opposite (charges) attract

1

potassium is more reactive (than hydrogen)
*accept potassium ions are less easily discharged (than hydrogen)
or potassium ions are less easily reduced (than hydrogen)*

1

(c) (i) gain electron(s)
*accept fully balanced correct equation for **2** marks*

1

one electron
*if no other marks awarded allow (potassium ions) reduced for
1 mark*

1

(ii) $2 \text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
*must be completely correct, including charge on electron
accept correct multiples*

1

(iii) 2, 8, 8
*accept any combination of dots, crosses, "e" or any other relevant
symbol
ignore any charges if given*

1

[10]

6.

(a) mixture has a lower melting point (than aluminium oxide)
*allow cryolite lowers melting point (of aluminium oxide)
ignore boiling point
do **not** accept cryolite is a catalyst*

1

(so) less energy needed
ignore cost

1

- (b) aluminium ions gain electrons 1
- (c) $2 \text{O}^{2-} \rightarrow \text{O}_2 + 4 \text{e}^-$
allow multiples
allow 1 mark for an unbalanced equation containing correct species 2
- (d) the electrode reacts with oxygen 1
- the electrode is carbon / graphite 1
- (so) carbon dioxide is produced
allow (so) the electrode / carbon / graphite is used up
allow (so) the electrode / carbon / graphite is burned away
ignore (so) the electrode / carbon / graphite is worn away
ignore (so) the electrode / carbon / graphite is corroded 1
- (e) *an answer of 941 (kg) scores 4 marks*
- (M_r of Al_2O_3 =) 102
- $\left(\frac{2\,000\,000}{102} =\right) 19\,608 \text{ (mol Al}_2\text{O}_3\text{)}$
allow correct calculation using incorrectly calculated value of M_r of Al_2O_3 1
- $\left(19\,608 \times \frac{3}{2} =\right) 29\,412 \text{ (mol O}_2\text{)}$
allow correct calculation using incorrectly calculated value of moles of Al_2O_3 1
- $\left(\frac{29\,412 \times 32}{1000} =\right) 941 \text{ (kg)}$
allow 941.1764706 (kg) correctly rounded to at least 2 significant figures
allow correct answer using incorrectly calculated value of moles of O_2 1

alternative approach:

(2 M_r of Al_2O_3 =) 204 (1)

204 (kg of Al_2O_3) gives 96 (kg of O_2) (1)

(2000 kg of Al_2O_3 gives)

$$\frac{2000}{204} \times 96 \text{ (kg of } \text{O}_2\text{)}$$

or

$$\frac{2000000}{204} \times 96 \text{ (g of } \text{O}_2\text{)} \text{ (1)}$$

$$= 941 \text{ (kg)} \text{ (1)}$$

(f) hydrogen (gas) would be produced (instead of sodium)

1

(because) sodium is more reactive than hydrogen

1

(g)

an answer of 50700 (dm^3) scores 2 marks

an answer of 50.7 (dm^3) scores 1 mark

$$\left(\frac{150\,000}{71} = \right) 2113 \text{ (mol of } \text{Cl}_2\text{)}$$

1

or

$$\text{(volume of 1 g of } \text{Cl}_2 = \frac{24}{71} =) 0.34 \text{ (dm}^3\text{)}$$

$$\left(\frac{150\,000}{71} \times 24 \right) = 50700 \text{ (dm}^3\text{)}$$

allow 50704.22535 (dm^3) correctly rounded to at least 2 significant figures

allow correct calculation using their calculated number of moles and/or calculated volume of 1 g

1

[16]

Examiner reports

1.

- (a) Most students were able to express the idea that activation energy is the minimum energy required to start a reaction. One common error was to say that it is the energy needed to make particles collide.
- (b) A variety of approaches was seen. The majority of students calculated the number of moles of both aluminium and of iron oxide and then attempted with varying degrees of success to explain why the number of moles of aluminium was insufficient for the iron oxide. Common errors were to omit to convert the mass from kilograms to grams, and/or using 54 as the ' M_r ' of aluminium. Of those who did correctly calculate the number of moles of each reactant, some had difficulty in expressing clearly the idea of 2 moles of aluminium being needed to react with 1 mole of iron oxide, and therefore 37.0 moles of aluminium is less than the 37.5 moles required, or that 18.75 moles of iron oxide is more than the 18.5 moles required. Other students started with the mass of one reagent and calculated the required mass of the other, either by the moles method or by proportion. These students found it easier to express why there was insufficient aluminium or excess iron oxide.
- (c) Few students scored both marks. Aqueous zinc was often seen as a response. Many students did not understand the idea of an ionic equation and tried to include the sulfate ion.
- (d) Only a third of students scored both marks, but a lot of students were confused about which species gained electrons and which lost them. Definitions of oxidation and reduction in terms of gain and loss of oxygen were not appropriate here and were not credited.

2.

- (a) 42% of students could answer this correctly. A hazard was asked for, and it was expected that students would use the correct chemical term for such a hazard; hence the idea that chlorine or carbon monoxide is toxic was required. Many did not use the correct term, referring instead to it being harmful (incorrect – it can kill, not merely make one ill), or dangerous or deadly (in what way?). Some students misread the equation and referred to carbon dioxide.
- (b) 40% of students achieved the mark here. Many did not realise that their observations of a vigorous reaction with a small piece of sodium in the lab would be magnified enormously on an industrial scale. Hence the reaction would be **very** violent or explosive, not just vigorous.
- (c) There were many good attempts at this question, with 73% of students scoring at least one mark. Many students referred to the lack of reactivity of the metals with argon, but some forgot to refer to air. Then some students did not express themselves clearly, referring to one or both metals reacting with air in general, or even with gases not present in the air, such as hydrogen.
- (d) There were some excellent attempts at this difficult question, which discriminated well. 8% of students achieved full marks with 43% achieving at least one mark. Full credit could be gained by those who started with the premise that titanium chloride must be ionic so should have a high melting point; very few started from the true situation, which is that titanium chloride must exist as small molecules as its melting point is low.

The mark that was scored least often was for the explanation of the bonding and how it leads to the actual, or predicted, melting point. Some spoiled an otherwise good description of ionic bonding by also referring to 'strong intermolecular forces'. Many referred to 'strong electrostatic forces' but failed to mention that those forces were between ions.

A significant number of students answered solely in terms of the separate elements, particularly titanium, rather than the compound.

- (e) 51% of students achieved the mark for this question. Whilst the definition of oxidation in terms of electrons is well known, the question asked why this reaction is oxidation. So, it was necessary to refer to the sodium atoms losing electrons rather than giving a generalised answer. Some students referred to 'it' losing electrons, without specifying what 'it' referred to, while others referred to oxygen, which is not present.
- (f) 54% of students achieved the mark here. It was not uncommon to see ' $-e^-$ ' on the right hand side. A significant number of incorrect responses showed the formation of a doubly charged sodium ion.
- (g) This difficult unstructured calculation tested a new area of the specification, and discriminated very well. 74% of students were able to achieve at least one mark, usually for calculating the M_r of $TiCl_4$. 7% of students achieved all four marks, with 12% achieving at least three marks and there were some valiant efforts.

Common errors were:

- using 92 instead of 23 to calculate the number of moles of Na
- omitting the unit conversion from kg to g for the mole calculations
- using the 1:4 mole ratio incorrectly or not at all.

Also, it is not appropriate to leave answers in the form of a fraction, even if this is the default on their calculator. Answers should always be converted to decimal form.

- (h) This was well answered, with 74% of students achieving both marks. A few forgot the 100 in their expression and didn't realise they were making about 1000 kg more than the theoretical maximum.

4.

- (a) Students found it difficult to score both marks on this question, although 77% of students scored one or more marks. Lower-attaining students simply stated that the electrolyte was a solid, without further explanation. Others incorrectly referred to the movement of delocalised electrons, rather than ions. Some referred to zinc instead of zinc chloride.
- (b) Students struggled to put sufficient detail into their answers, although no more was required than what is directly stated in the specification. 9% of students achieved all three marks.

A lot of weakly expressed ideas were seen. Instead of stating that each atom forms three covalent bonds, many students stated that graphite has three covalent bonds, or that three out of four carbon atoms are bonded, both of which are untrue.

Very few said that each carbon atom has one delocalised electron: a statement directly from the specification. Those who did score the marks for the structure often didn't say that delocalised electrons move or carry charge **through** the structure. They often implied the delocalised electrons just moved around at random. Many students misunderstood the process of electrical conductivity completely, thinking that it is something to do with layers sliding over each other.

Some students referred to other properties such as softness, instead of confining their answer to the question posed.

- (c) 58% of students did not achieve a mark for this question. The investigation was stated to be about the relationship between volume and time, but the diagram showed no means of measuring volume. The expected answer was to use inverted measuring cylinders so that volume (not 'amount') could be measured.

Although it would pose practical difficulties, the idea of using gas syringes was also accepted (on the grounds that it could be made to work if they were attached to funnels placed over the electrodes). Many suggested changing the electrodes as graphite is not inert, suggesting a lack of understanding of the required practical. Some students suggested that the apparatus is upside down.

- (d) Although 75% of students achieved one mark, usually for the trend in the hydrogen graph, just 7% achieved three marks.

In order to score the mark for proportionality of the hydrogen graph, both variables had to be stated, not just 'it is proportional'. Alternatively, a steady rate of collection would gain the mark. It had not been anticipated that students would calculate the rate of collection, but those who did were awarded credit. Some referred to proportionality correctly, but did not state which gas they were referring to.

The chlorine line posed more difficulty. Many described the chlorine line after 8 minutes as proportional. It is not as it shows a linear relationship, where the volume of chlorine increases at a steady rate.

Students are very unsure of how to describe trends in curved graphs; the idea of the chlorine being collected at an increasing rate (which could be expressed as faster and faster) being very rarely seen.

As in other questions, the language used by students created a barrier to attaining marks. 'It remains constant' could mean the gas volume no longer changes, rather than the rate of collection.

- (e) 2% of students were able to answer this question. Many students either had not read or understood the statement that the number of moles of each gas produced is the same. There were attempts to justify the difference in terms of reactivity, formula mass, size of molecules, pressure difference or the concentration of H^+ and Cl^- in solution. Some students referred to the length of time taken for the chlorine to react, rather than to be formed. Others suggested the molar volume was different for different gases, ignoring what they know from the part of the specification on gas volumes.

The formation of chlorine and hydrogen is part of the electrolysis required practical, so students should not be completely unfamiliar with the bubbles of chlorine taking longer to form. They should also be familiar with 'chlorine water' from their work on Group 7.

- (f) Students found this calculation on a new area of the specification quite difficult, with 26% of students providing a fully correct answer.

It was necessary to read off the correct volume from the graph and change the units from cm^3 to dm^3 , or to change those of the gas molar volume to cm^3/mol , but many students didn't do this. Many students also struggled to express their answer in standard form.

5.

- (a) (i) Many students referred to "potassium" rather than a potassium salt. While over half of the students realised either that the solid salt was unable to conduct electricity or that the salt should be molten or in solution to conduct, fewer students indicated the necessary movement of particles to gain the second marking point. A number of students thought that electrons moved through the electrolyte. A small minority of students answered in terms of the reactivity of potassium.
- (ii) There was a wide variety of incorrect responses, often focusing on the reactivity of potassium, or that it was a new and useful metal, or that this method could be used to produce other elements. The answers of the minority of students who gained credit often related to the ideas of Davy having status and Davy having proof or evidence. A small number of students did not give any added value in their response, stating that he was a professor.
- (b) The most able students gave succinct and accurate responses, gaining full credit. Carbon dioxide and chlorine were the commonly mentioned incorrect gases, and students giving these gases had difficulty answering the rest of the question. Answers from weaker students indicated a very poor understanding of basic electrolysis, and therefore no grasp of the relevance of reactivity on the discharge of ions. Some students omitted to mention that hydrogen ions are positive.
- (c) (i) Students are more familiar with describing how ions are formed from atoms, so this question proved to be discriminating as it required more thought, and a minority of students earned both marks. Some students explained why potassium ions are attracted to the negative electrode rather than how they form potassium atoms. A number of students thought that the potassium atom would have a full outer shell of electrons.
- (ii) Although a minority, an encouraging number of students gained this mark. The half equation had to be completely correct, including the correct charge on the electrons to gain credit. Incorrect responses often included potassium atoms or ions.
- (iii) Approximately half of students gained credit, giving a clearly drawn electronic structure of a chloride ion. A chlorine atom structure (2,8,7) was the most commonly seen incorrect response, while some students gave only 6 electrons on the outer shell, suggesting that they thought that a chlorine atom would lose, rather than gain an electron on ionisation.

6.

- (a) The explanation for the use of cryolite in the extraction of aluminium does not appear to be well known. Almost two-thirds of students scored 0 marks.

A common misconception was that cryolite acts as a catalyst.

Common responses seen that were not credited included:

- to melt the aluminium oxide so that it conducts electricity
- so that the ions can move freely
- to reduce costs.

- (b) Almost 60% of students correctly identified that aluminium ions gain electrons as the process occurring at the negative electrode.
- (c) Students found the completion of the half-equation difficult. Almost 70% scored 0 marks.

Common errors included:

- including Al / Al_2O_3
- representing the oxide ion as O^- or O_2^-
- 2 electrons rather than 4 electrons being produced.

- (d) However, only approximately a quarter of students were able to state why the positive electrode needs to be replaced and hence score all 3 marks.
- (e) Most students scored at least 1 mark. Approximately 40% scored all 4 marks.

The most frequently scored mark was for $M_r(\text{Al}_2\text{O}_3) = 102$ (or '204' from the alternative approach in the mark scheme).

A common error was $M_r(\text{Al}_2\text{O}_3) = 156$ but students were still able to score marks for using this incorrect value in subsequent steps.

A valid variation on the reacting mass method was to calculate the percentage by mass of oxygen in Al_2O_3 and then use this to calculate the mass of oxygen in 2000 kg of Al_2O_3

It was not always clear which approach was being used. A common incorrect final answer was 1412 (kg) where students had applied the 2 : 3 mole ratio (from the main approach) as well as the 96 / 204 mass ratio (from the alternative approach).

- (f) Students found this question difficult. Almost two-thirds of students scored 0 marks.

Even when students realised that reactivity was the key idea, they sometimes compared the reactivity of the wrong species, for example, sodium and chlorine or Na^+ ions and H^+ ions

A common misconception was that sodium chloride solution does not conduct electricity.

- (g) Approximately two-thirds of students scored at least 1 mark for this calculation.

However, the most common final answer was 50.7 dm^3 . The error here was the omission of the unit conversion, kg to g.